

COMMENT ON “CONTINUITY OF THE DISTRIBUTION FUNCTION OF THE ARGMAX OF A GAUSSIAN PROCESS”: ABSOLUTE CONTINUITY AND ANTI-CONCENTRATION

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ABSTRACT. Cattaneo, Cox, Jansson, and Nagasawa (2026) prove continuity of the multivariate distribution function of a Gaussian argmax. Under the same assumptions, this Comment proves that the joint law has a Lebesgue density belonging locally to L^p for every $1 \leq p < \infty$. It also gives explicit bounds for Borel sets and anti-concentration inequalities for linear contrasts. The proof combines translation averaging, Cameron–Martin tilting, and localization. The bounds control proximity to regular indifference restrictions and, when combined with approximation estimates, the sensitivity of economic decisions to resampling or computational error.

1. INTRODUCTION

Many nonregular estimators maximize a locally nonsmooth criterion, and their normalized estimation error often converges to

$$r_n(\widehat{\theta}_n - \theta_0) \rightsquigarrow \widehat{S} := \arg \max_{s \in \mathbb{R}^d} G(s), \quad (1)$$

where $r_n \rightarrow \infty$, $\rightsquigarrow$ denotes convergence in distribution, and G is a nonstationary Gaussian process. Limits of the form (1) appear in cube-root asymptotics (Kim and Pollard, 1990), maximum score (Manski, 1975), empirical risk minimization (Mohammadi and van de Geer, 2005), and threshold models (Hansen, 2000; Yu and Fan, 2021). Ordinary bootstrap inference can fail in these settings (Abrevaya and Huang, 2005), motivating tailored resampling and numerical procedures (Patra et al., 2018; Cattaneo et al., 2020; Hong and Li, 2020).

Keywords. Argmax distribution, cube-root asymptotics, Gaussian process, stationary increments, Cameron–Martin theorem, anti-concentration.

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Cattaneo et al. (2026) give high-level conditions under which the joint cumulative distribution function of $\widehat{S}$ is continuous. This excludes atoms in each coordinate marginal and supports quantile-based inference. It does not, however, preclude concentration on a diagonal or another lower-dimensional restriction. Such restrictions are economically natural: they describe equal coefficients, zero marginal effects, welfare-neutral policy changes, and regime indifference.

Without strengthening the assumptions of Cattaneo et al. (2026), this Comment proves that $\widehat{S}$ has a joint Lebesgue density belonging locally to every finite L^p . It also gives explicit probability bounds for arbitrary Borel sets and anti-concentration inequalities for every contrast $a^\top \widehat{S}$, with a polynomial rate in self-similar models. The gain is therefore from coordinatewise atomlessness to joint and quantitative regularity. The result strengthens, rather than corrects, Cattaneo et al. (2026).

Stationary increments yield a translation-averaged volume bound for the centered compact argmax. Cameron–Martin tilting transfers it to the drifted process, and localization recovers the global maximizer. No path differentiability is needed. Earlier location-density results include stationary processes (Samorodnitsky and Shen, 2013) and smooth Gaussian fields (Azaïs and Chassan, 2020). Aurzada and Helmer (2026, Theorem 2.4) prove an interior-density theorem for compact argmaxes of stationary-increment processes. Combined with local Cameron–Martin equivalence and localization, that theorem already implies qualitative absolute continuity in our setting. Our purpose is to make the strengthening quantitative: we give a direct Borel-set estimate with explicit constants, local L^p bounds, and moduli for economic contrasts. Unlike anti-concentration bounds for Gaussian maximum *values* (Chernozhukov et al., 2015), our inequalities concern the maximizing *location*.

2. MAIN RESULT

Fix $d \geq 1$, write $Q_r = [-r, r]^d$, and let $\mathcal{L}^d$ denote Lebesgue measure. The norms $\|\cdot\|_2$ and $\|\cdot\|_\infty$ are the Euclidean and maximum norms, and Φ is the standard normal distribution function. Let G be a real-valued Gaussian process on $\mathbb{R}^d$ with mean μ and covariance C . Put $X = G - \mu$, and let $\mathcal{H}_R$ be the reproducing-kernel Hilbert space (RKHS) of $C|_{Q_R \times Q_R}$. Here

$C(D)$ denotes the real continuous functions on D , with uniform convergence on compact sets (the uniform norm when D is compact).

Assumption 2.1 (Cattaneo–Cox–Jansson–Nagasawa). (i) *Almost surely, G has continuous paths and attains its maximum on $\mathbb{R}^d$.*

(ii) *For all $s, t \in \mathbb{R}^d$ and $h \in \mathbb{R}^d \setminus \{0\}$, $C(h, h) > 0$ and*

$$C(h + s, h + t) - C(h + s, h) - C(h, h + t) + C(h, h) = C(s, t). \quad (2)$$

(iii) *For every integer $N \geq 1$, $\mu|_{Q_N} \in \mathcal{H}_N$.*

Part (i) supplies path continuity and global attainment, part (ii) implies compact uniqueness and translation geometry for the centered field, and part (iii) permits a local change of measure.

Proposition 2.2. *Under Assumption 2.1, μ is continuous, C is jointly continuous, $\mu(0) = 0$, $X(0) = G(0) = 0$ almost surely, X has stationary increments, and*

$$\text{Var}(X(s) - X(t)) = C(s - t, s - t) > 0 \quad (s \neq t). \quad (3)$$

On each fixed nonempty compact set, the maximizers of X and G are almost surely unique; the attained global maximizer of G is also almost surely unique. Moreover, $\mu|_{Q_R} \in \mathcal{H}_R$ for every $R > 0$. Writing

$$P_R^0 = \text{Law}(X|_{Q_R}), \quad P_R = \text{Law}(G|_{Q_R}), \quad K_R = \|\mu|_{Q_R}\|_{\mathcal{H}_R}, \quad L_R = \frac{dP_R}{dP_R^0},$$

the path laws are equivalent and, for $1 < p < \infty$,

$$\|L_R\|_{L^p(P_R^0)} = \exp\left(\frac{p-1}{2} K_R^2\right). \quad (4)$$

Lemma 2.3. *For each nonempty compact D , there is a Borel argmax selector on $C(D)$. There is also a Borel map on $C(\mathbb{R}^d)$ that agrees with the global argmax whenever it is unique and attained.*

The standard verifications are recorded in Appendix A. Let

$$\widehat{S} = \arg \max_{s \in \mathbb{R}^d} G(s), \quad \tau(r) = \mathbb{P}(\widehat{S} \notin Q_r),$$

so $\tau(r) \downarrow 0$. For $K \geq 0$, define

$$\Psi_K(u) = \Phi(\Phi^{-1}(u) + K), \quad 0 < u < 1, \quad \Psi_K(0) = 0, \quad \Psi_K(1) = 1.$$

Theorem 2.4 (Density and anti-concentration). *Under Assumption 2.1:*

(i) *for $0 < r < R$ and every Borel set $A \subseteq Q_r$,*

$$\mathbb{P}(\widehat{S} \in A) \leq \Psi_{K_R} \left(\min \left\{ 1, \frac{\mathcal{L}^d(A)}{(R-r)^d} \right\} \right). \quad (5)$$

Consequently, $\text{Law}(\widehat{S}) \ll \mathcal{L}^d$, and its density satisfies, for $1 \leq p < \infty$,

$$\|f_{\widehat{S}}\|_{L^p(Q_r)} \leq \inf_{R>r} (R-r)^{-d(1-1/p)} \exp \left(\frac{p-1}{2} K_R^2 \right); \quad (6)$$

(ii) *for $a \neq 0$ and $\varepsilon > 0$,*

$$\sup_{t \in \mathbb{R}} \mathbb{P}(|a^\top \widehat{S} - t| \leq \varepsilon) \leq \inf_{0 < r < R} \left[\tau(r) + \Psi_{K_R} \left(\min \left\{ 1, \frac{2^d r^{d-1} \varepsilon}{(R-r)^d \|a\|_\infty} \right\} \right) \right]. \quad (7)$$

The right-hand side converges to zero as $\varepsilon \downarrow 0$.

The bounds are process-specific through K_R and $\tau(r)$; the assumptions alone do not supply a common rate over a class of models.

Corollary 2.5 (Self-similar fields). *Suppose in addition that there are $H > 0$, $\beta > H$, and $c_0 > 0$ such that, for every $\rho > 0$,*

$$(X(\rho s))_{s \in \mathbb{R}^d} \stackrel{\text{fdd}}{=} (\rho^H X(s))_{s \in \mathbb{R}^d}, \quad \mu(\rho s) = \rho^\beta \mu(s), \quad \mu(s) \leq -c_0 \|s\|_\infty^\beta.$$

Here $\stackrel{\text{fdd}}{=}$ denotes equality of finite-dimensional distributions. Set

$$0 < \sigma^2 = \sup_{1 \leq \|s\|_\infty \leq 2} C(s, s), \quad \eta_* = \left(\frac{c_0}{c_0 + \sigma K_1} \right)^2.$$

For every $0 < \eta < \eta_*$, there are $C_\eta < \infty$ and $\varepsilon_\eta > 0$ such that

$$\sup_{\|a\|_2=1} \sup_{t \in \mathbb{R}} \mathbb{P}(|a^\top \widehat{S} - t| \leq \varepsilon) \leq C_\eta \varepsilon^\eta, \quad 0 < \varepsilon < \varepsilon_\eta. \quad (8)$$

The sufficient exponent compares coercivity with the product σK_1 ; no optimality or endpoint claim is made. These quantities need not vary independently: multiplying μ by a positive constant scales both c_0 and K_1 , leaving η_* unchanged, although the concentration constants and spatial scale may change.

The distinction from Theorem 1 of Cattaneo et al. (2026) is substantive: if Z is continuously distributed, (Z, Z) has a continuous joint distribution function but lies on a diagonal. Theorem 2.4 rules out all Lebesgue-null restrictions, not just coordinate atoms, and quantifies small-set probabilities.

Keeping continuity and global attainment, density existence needs only unique compact argmaxes with translation-equivariant locations and one-sided local domination of the perturbed path law. The explicit bounds additionally use control of the likelihood ratio. In a fixed-covariance Gaussian translation family, however, such domination is equivalent to local Cameron–Martin membership. Thus the argument extends abstractly beyond full stationary increments, but it does not conceal a weaker Gaussian drift condition.

3. ECONOMIC INTERPRETATION

In the main applications, $\mu(0) = 0$ and $\mu(s) \leq 0$: $-\mu(s)$ is the population loss from departing from the identified action, while X records random local criterion differences. Translation equivariance expresses homogeneity of these differences, not stationarity of payoff levels. The factor $(R - r)^{-d}$ reflects that geometry. The relative entropy $D_{\text{KL}}(P_R \| P_R^0) := \mathbb{E}_{P_R} \log L_R = K_R^2/2$ measures local change-of-measure distortion, whereas $\tau(r)$ measures failure of localization. These statements concern the limiting estimation error, not randomness of the population parameter or finite-sample density existence.

Indifference sets. If $M \subset \mathbb{R}^d$ is Borel and $\mathcal{L}^d(M) = 0$, then $\mathbb{P}(\widehat{S} \in M) = 0$. For a compact embedded C^1 submanifold $N \subseteq Q_r$ of codimension $k \geq 1$ (possibly with boundary), define

$$N_\varepsilon = \{s \in Q_r : \inf_{z \in N} \|s - z\|_2 \leq \varepsilon\}.$$

A finite Lipschitz-chart cover gives $\mathcal{L}^d(N_\varepsilon) = O(\varepsilon^k)$. Hölder's inequality and (6) give $\mathbb{P}(\widehat{S} \in N_\varepsilon) = O(\varepsilon^{k\alpha})$ for every $0 < \alpha < 1$. This applies to coefficient or welfare restrictions when their zero sets are regular manifolds; an arbitrary indifference set need not be null.

Approximation and decisions. For $Y = a^\top \widehat{S}$ and a bootstrap, simulated, or numerical approximation $\widetilde{Y}$ defined on the same probability space, and for every $\delta > 0$,

$$\sup_t |\mathbb{P}(\widetilde{Y} \leq t) - \mathbb{P}(Y \leq t)| \leq \mathbb{P}(|\widetilde{Y} - Y| > \delta) + \sup_t \mathbb{P}(|Y - t| \leq \delta). \quad (9)$$

In (9), the first term is approximation error and the second is the near-indifference probability controlled by (7). In particular, a sign or policy ranking can change only when the approximation error is large or the ideal statistic lies near the relevant decision boundary.

For a finite policy menu with distinct affine payoff contrasts, a union bound controls near-ties after each nonzero contrast is normalized. Identical payoffs are not covered by this argument. Confidence-region perturbations similarly require control of the corresponding boundary layer. In every case the theorem supplies only the anti-concentration component; the displacement bound is procedure-specific. For a conditional bootstrap conclusion, the coupling and the target marginal must be justified conditionally on the data. The need for tailored resampling also arises in related shape-constrained problems (Cattaneo et al., 2024); regularity of the limiting law alone does not resolve that problem.

Let us indicate some applications of our main result.

Maximum score. Manski's estimator selects the index rule that best matches observed binary choices. For a fixed nonzero covariate contrast x in the identified coordinates, $x^\top \widehat{S}$ is the limiting perturbation of the latent-utility index. Its concentration bound controls sensitivity of predictions near the population decision boundary.

Empirical risk minimization. Discontinuous classification losses can yield Brownian-type argmax limits. In the separable example of Cattaneo et al. (2026, Section 2.2), the coordinates already have independent scaled Chernoff laws (Chernoff, 1964; Groeneboom, 1989); density existence is not new there. The theorem provides a common quantitative bound and controls near-risk ties whenever they correspond to regular restrictions on the limiting optimizer.

Threshold regression. In the models with shrinking regime effects of Cattaneo et al. (2026, Section 2.3), localization reflects regime separation and the contrast bound controls the limiting boundary error. This Gaussian argument does not cover every fixed-jump threshold asymptotic experiment.

4. PROOF

For $R > 0$, write $M_R^X = \arg \max_{s \in Q_R} X(s)$. The geometric core is the following translation-averaging estimate.

Lemma 4.1 (Translation averaging). *For $0 < r < R$ and Borel $A \subseteq Q_r$,*

$$\mathbb{P}(M_R^X \in A) \leq \frac{\mathcal{L}^d(A)}{(R-r)^d}. \quad (10)$$

Hence $\text{Law}(M_R^X)|_{Q_r}$ has a density bounded by $(R-r)^{-d}$.

Proof. Set $H = Q_{(R-r)/2}$, $D = Q_R$, and $D_* = \bigcap_{h \in H} (D + h) = Q_{(R+r)/2}$. For each fixed h , the processes $X(h + \cdot) - X(h)$ and X have the same law on $C(D)$: stationary increments and evaluation on a countable dense set establish this equality. Subtracting $X(h)$ leaves the argmax unchanged, so compact uniqueness gives

$$\arg \max_{s \in D+h} X(s) - h \stackrel{d}{=} M_R^X, \quad h \in H. \quad (11)$$

Also $A + h \subseteq D_*$: for $a \in A$ and $h, k \in H$, $a + h - k \in Q_r + Q_{R-r} = D$, so $a + h \in D + k$.

Let $p = \mathbb{P}(M_R^X \in A)$. By (11),

$$p = \mathbb{P}\left(\arg \max_{s \in D+h} X(s) \in A + h\right) \leq \mathbb{P}\left(\arg \max_{s \in D_*} X(s) \in A + h\right), \quad h \in H.$$

Indeed, on the event in the middle, the maximizer over $D + h$ lies in $A + h \subseteq D_*$ and is therefore the unique maximizer over $D_* \subseteq D + h$. Write $M_* = \arg \max_{D_*} X$. By Lemma 2.3, $(h, \omega) \mapsto M_*(\omega) - h$ is measurable, so Tonelli's theorem yields

$$\mathcal{L}^d(H)p \leq \mathbb{E} \int_H \mathbf{1}\{M_* \in A + h\} dh \leq \mathcal{L}^d(A),$$

since, for fixed z , the admissible shifts form $H \cap (z - A)$. As $\mathcal{L}^d(H) = (R - r)^d$, (10) follows; the density statement is its Radon–Nikodym consequence. $\square$

Lemma 4.2 (Sharp Gaussian transfer). *Let P_0 be a centered Gaussian measure on a separable Banach space, and let P be its translate by a Cameron–Martin vector of norm K . For every Borel event E ,*

$$P(E) \leq \Psi_K(P_0(E)). \quad (12)$$

Moreover, for $0 < \alpha < 1$ and $0 \leq u \leq 1$,

$$\Psi_K(u) \leq \exp\left(\frac{\alpha K^2}{2(1-\alpha)}\right) u^\alpha. \quad (13)$$

Proof. For $K = 0$, (12) is equality and (13) is $u \leq u^\alpha$. The probability endpoints follow from equivalence. Suppose $K > 0$ and $u = P_0(E) \in (0, 1)$. Under P_0 , the likelihood ratio has the form

$$L := \frac{dP}{dP_0} = \exp\left(KZ - \frac{K^2}{2}\right), \quad Z \sim N(0, 1).$$

Let $c = \Phi^{-1}(1 - u)$, $H_c = \{Z \geq c\}$, and $\ell_c = \exp(Kc - K^2/2)$. Since $P_0(H_c) = u$,

$$\mathbb{E}_{P_0}[L(\mathbf{1}_E - \mathbf{1}_{H_c})] = \mathbb{E}_{P_0}[(L - \ell_c)(\mathbf{1}_E - \mathbf{1}_{H_c})] \leq 0.$$

The last integrand is nonpositive both on H_c and on its complement. Completing the square in the Gaussian density gives

$$\mathbb{E}_{P_0}[L\mathbf{1}_{H_c}] = \int_c^\infty e^{Kz - K^2/2} d\Phi(z) = \int_c^\infty d\Phi(z - K) = \Phi\left(\Phi^{-1}(u) + K\right),$$

which proves (12), with equality for $E = H_c$. For $0 < \alpha < 1$, set $q = (1 - \alpha)^{-1}$. Applying Hölder’s inequality to this extremal event gives

$$\Psi_K(u) = \mathbb{E}_{P_0}[L\mathbf{1}_{H_c}] \leq \|L\|_{L^q(P_0)} u^{1-1/q} = \exp\left(\frac{\alpha K^2}{2(1-\alpha)}\right) u^\alpha.$$

This proves (13) for all u , including the endpoints by continuity. $\square$

Proof of Theorem 2.4. Let $T_R : C(Q_R) \rightarrow Q_R$ be the selector from Lemma 2.3, and set $M_R^G = T_R(G|_{Q_R})$. If $\widehat{S} \in A \subseteq Q_r$, then $M_R^G = \widehat{S} \in A$. Under the centered path law,

Lemma 4.1 bounds the same Borel event $\{T_R \in A\}$ by

$$u_A = \min \left\{ 1, \frac{\mathcal{L}^d(A)}{(R-r)^d} \right\}.$$

Under the path law of G , Lemma 4.2 with norm K_R gives (5). Null sets and an exhaustion by cubes prove joint absolute continuity.

For (6), fix $1 < p < \infty$ and let ν_0 and ν be the laws of T_R under P_R^0 and P_R . Then $\nu \ll \nu_0$; put $g = d\nu/d\nu_0$. On Q_r , write $d\nu_0 = f_0 dx$, where $f_0 \leq (R-r)^{-d}$. The defining integral identity for g gives

$$g(T_R) = \mathbb{E}_{P_R^0}[L_R \mid \sigma(T_R)], \quad \int g^p d\nu_0 \leq \mathbb{E}_{P_R^0} L_R^p$$

by conditional Jensen. Consequently,

$$\int_{Q_r} (g f_0)^p dx = \int_{Q_r} g^p f_0^{p-1} d\nu_0 \leq (R-r)^{-d(p-1)} \mathbb{E}_{P_R^0} L_R^p.$$

The global argmax law restricted to Q_r is dominated by ν ; hence its density is bounded almost everywhere by $g f_0$. Together with (4), this proves (6) for $p > 1$. For $p = 1$, the restricted global law is a subprobability measure.

Finally, the slab $A_{r,t,\varepsilon} = \{s \in Q_r : |a^\top s - t| \leq \varepsilon\}$ satisfies

$$\mathcal{L}^d(A_{r,t,\varepsilon}) \leq \frac{2^d r^{d-1} \varepsilon}{\|a\|_\infty}, \quad (14)$$

by slicing along a coordinate of maximal $|a_j|$. Splitting according to $\widehat{S} \in Q_r$, inserting (14) into (5), and taking the infimum over $0 < r < R$ proves (7). Its right-hand side vanishes by first choosing r large and then letting $\varepsilon \downarrow 0$ with $R > r$ fixed. $\square$

Thus local density control comes from translation and tilting, whereas unbounded contrasts also require localization. None of the bounds asserts a locally bounded density or continuity of that density.

APPENDIX A. STANDARD CONSEQUENCES OF THE ASSUMPTIONS

Proof of Proposition 2.2. Putting $s = t = 0$ in (2) gives $C(0,0) = 0$; positive semidefiniteness then gives $C(s,0) = 0$, so the identity also holds for $h = 0$. Gaussianity yields

stationary increments and (3); see Cattaneo et al. (2026, Lemma 1). Sample continuity gives continuity of the Gaussian mean and covariance. The reproducing identity implies $\mu(0) = 0$, hence $X(0) = G(0) = 0$ almost surely. Compact uniqueness follows from Kim and Pollard (1990, Lemma 2.6); a countable cube exhaustion gives global uniqueness. RKHS restriction (Aronszajn, 1950) gives $\mu|_{Q_R} \in \mathcal{H}_R$, and the Cameron–Martin theorem (Giné and Nickl, 2016, Theorem 2.6.13) gives local equivalence and (4). $\square$

Proof of Lemma 2.3. For nonempty compact D , the measurable maximum theorem (Aliprantis and Border, 2006, Theorem 18.19) supplies a Borel selection from the argmax correspondence on $C(D)$; on the uniqueness locus it is the maximizer. Apply these selectors to the cubes (Q_n) . On the Borel set where the resulting sequence converges, take its limit, and set the selector equal to 0 otherwise. For a continuous path with a unique attained global maximum, the sequence is eventually constant at that maximizer. $\square$

APPENDIX B. POLYNOMIAL REFINEMENT

Proof of Corollary 2.5. Put $\gamma = 2(\beta - H)$. The map $h \mapsto r^{-H}h(r \cdot)$ is an isometry from $\mathcal{H}_r$ onto $\mathcal{H}_1$ by covariance scaling (Aronszajn, 1950). Homogeneity of μ therefore gives

$$K_r = r^{\beta-H} K_1. \quad (15)$$

Let $\mathcal{A} = \{s : 1 \leq \|s\|_\infty \leq 2\}$ and $m = \mathbb{E} \sup_{s \in \mathcal{A}} X(s) < \infty$ by Fernique’s theorem. If $\widehat{S} \notin Q_r$, then, for some $k \geq 0$, $2^k r \leq \|\widehat{S}\|_\infty \leq 2^{k+1} r$ and $X(\widehat{S}) \geq -\mu(\widehat{S}) \geq c_0(2^k r)^\beta$, because $G(\widehat{S}) \geq G(0) = 0$. After rescaling this shell to $\mathcal{A}$, the required excursion level is $c_0(2^k r)^{\beta-H}$. Fix $0 < b < c_0^2/(2\sigma^2)$ and choose $\theta \in (0, 1)$ with $b < \theta^2 c_0^2/(2\sigma^2)$. For large r , $m \leq (1 - \theta)c_0 r^{\beta-H}$, so the Borell–Tsirelson–Ibragimov–Sudakov inequality (Adler and Taylor, 2007, Theorem 2.1.1) bounds the k th shell by $\exp\{-c(2^k r)^\gamma\}$ for some $c > b$. Since $2^{k\gamma} \geq 1 + k(2^\gamma - 1)$, summation gives

$$\tau(r) \leq A_b e^{-br^\gamma} \quad (16)$$

for all sufficiently large r .

Coercivity implies $K_1 > 0$. Fix $0 < \eta < \eta_*$. Choose $0 < b < c_0^2/(2\sigma^2)$ sufficiently close to the upper endpoint and $\lambda > 1$ sufficiently close to one so that

$$\alpha = \frac{\sqrt{2b}}{\lambda^{\gamma/2}K_1 + \sqrt{2b}} \quad \text{satisfies} \quad \alpha^2 > \eta.$$

Apply (13) to (7), use (15) and (16), take $R = \lambda r$, and use $\|a\|_\infty \geq d^{-1/2}$ for $\|a\|_2 = 1$. With $c_{d,\lambda} = 2^d \sqrt{d}/(\lambda - 1)^d$, the concentration function is bounded by

$$A_b e^{-br^\gamma} + e^{\alpha \lambda^\gamma K_1^2 r^\gamma / [2(1-\alpha)]} \left(\frac{c_{d,\lambda} \varepsilon}{r} \right)^\alpha.$$

Set

$$r^\gamma = q \log(1/\varepsilon), \quad q = \frac{2}{(\lambda^{\gamma/2}K_1 + \sqrt{2b})^2}.$$

For sufficiently small ε , this choice is admissible. Then $bq = \alpha^2$ and $\lambda^\gamma K_1^2 q/2 = (1 - \alpha)^2$. The first term is $A_b \varepsilon^{\alpha^2}$, and the second is

$$c_{d,\lambda}^\alpha q^{-\alpha/\gamma} [\log(1/\varepsilon)]^{-\alpha/\gamma} \varepsilon^{\alpha^2}.$$

The logarithmic factor is bounded for small ε . Since $\alpha^2 > \eta$, (8) follows. $\square$

Acknowledgments. The author would like to thank Professors Matias D. Cattaneo, Sijie Luo and Vladimir Sunski for helpful communications.

Data availability. No data were used or generated in this Comment.

Conflict of interest. The author declares no conflict of interest.

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